

Machine Learning-based Attack Detection System with AES Encryption for Secure CSV Data Processing

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Abstract - The current research introduces a convenient workflow on intrusion detection and data security integrating a neural-network-based binary classifier with a symmetric encryption module to promote secure data management in the networked setting. The system is built entirely by the implemented components of the project which are a Streamlit interface, a Keras-based trained model of ANN followed by a preprocessing and label-mapping utility and an AES-CBC pipeline of encryption and decryption. The TRAbID2017 data and the class-label file are utilized to train and test the classifier as well as, train behavior is analyzed by the visualization of accuracy and loss which are produced in the development notebook. The architecture helps customers to add Csv files, confirm the entries, get the predictions and safeguard sensitive data which is automatically encrypted before transmission or storage. Its design is also usable and transparent to offer decrypted and or encrypted file previews and prediction outputs via the application interface. The workflow will show how neural models based on lightweight computations and symmetric cryptographic functions can provide an analytical power and secure data handling without supposing anything to the project code. This contribution provides a vivid illustration of a mixed learning-and-encryption pipeline that can be applicable in the settings where the support of the data confidentiality should be combined with the support of the data attack detection.

Keywords

Neural Networks, intrusion detection, encryption, AES, Streamlit, and data processing, prediction pipeline, cybersecurity, data protection, and classification.

I. INTRODUCTION

Increased reliance on interdependent digital systems has heightened the urgency to have unified mechanisms that both facilitate in the effective detection of intrusion and the maintenance of uncompromising work on data. The contemporary network environments produce significant amounts of heterogeneous and, accordingly, manual monitoring becomes unfeasible, leaving the environment more vulnerable to manipulation of data, spoofing, ransomware, and targeted attacks. The recent researches regularly stress that the cyber-physical system, smart grids, cloud computing, car networks, and industrial control settings need security solutions that could detect malicious actions and protect the confidentiality and integrity of data shared at the same time [1]-[7]. With the help of the ever-developing encryption methods, the attackers are increasingly trying to interfere or bypass the cryptographic operations. Neural network-based machine learning has emerged as a popular method to be used in identifying such attacks because it can absorb subtle changes in network data behavior in high-dimensional data [2], [5], [9].

Various publications have covered unified or hybrid security used on data classification with some cryptographic protection. Ensemble and neural models have shown to be effective in detecting data integrity attacks in the smart grid setting [1], and hybrid encryption using deep learning has already been used to protect large-scale data sharing operations [4]. Equally, in intrusion detection systems of industrial control systems, drones and vehicular networks, as well as cloud settings, lightweight learning models and cryptographic protection are becoming inseparable to support resilient communication pipelines [3], [7],

[11], [12]. These tendencies indicate a positive tendency of detecting and encryption being not regarded as separate parts, but as elements of a larger cybersecurity strategy.

In this respect, the current paper strictly addresses the adopted elements of the project code: a neural-network binary classifier which is trained with the help of the TRAbID2017 dataset, a symmetric encryption module created based on the AES-CBC algorithm, and a workflow used by a person in the form of a Streamlit interface. The classifier assists in the detection of both benign and malicious entries in the traffic, and the encryption logic makes it possible to keep sensitive input files safely in storage, transmit, and process them. Similar studies have revealed that next-generation software and technical systems in cybersecurity will persist to adopt adversarial robustness, adaptive cryptography, and machine-learning-based detection [8], [10], [13]-[15]. Even though this piece of work is constrained by the implemented system components, it is consistent with the overall trend of integrating, machine-learning-focused security architecture.

Using the strict allocation of functions and actions that are present in the project files, this research paper will give a vivid, implementation-based illustration of how a neural classifier and a symmetric encryption mechanism can be integrated to facilitate a safe data assessment. The work is part of the existing work on creating practical, convenient, and computationally efficient security machines that can be applied in real world usage.

Scope of Detected Intrusions

The proposed system performs binary intrusion detection based on the labeling structure of the TRAbID2017 dataset. Rather than identifying individual attack categories, the classifier distinguishes between benign and malicious network traffic instances. The malicious class represents aggregated intrusion behaviors present in the dataset, enabling the system to identify abnormal or attack-related activity at a broader level. This design directly reflects the binary output configuration implemented in the neural network model used for training and inference.

II. LITERATURE SURVEY

The Intrusion detection machine learning.

The field of machine learning has now become a ubiquitous component of automated intrusion detection due to its capacity to represented intricate and high dimensional traffic patterns, and to identify subtle anomalies of the normal pattern. Overall surveys indicate that supervised and hybrid schemes are still popular in real-life applications of IDS with labeled data sets, particularly in applications where immediate decisions must be made [14]. Neural and ensemble classifier performance in domain-specific applications can be shown to be effective in industrial control and IoT applications, where the constraints in resource

usage and high-security requirements of operation increase pressure on detection errors and latency [3]. Tasks regarding drones and small platforms are based on lightweight models, and feature selection as priorities to balance both the detection performance and the computational cost, validating the usefulness of small neural designs in real-world IDS assignments [11].

Data protection and hybrid encryption.

Recent studies emphasize a multifaceted approach of the detection based on learning with cryptographic guardianship and this offers provision of situational knowledge and confidentiality of information. Hybrid encryption framework studies use symmetric and asymmetric parts to ensure massive sharing of data, and incorporating deep-learning modules to validate data integrity and implement access control enforcement [4]. In intelligent-grid and industrial applications, combined schemes with ensemble schemes prove to be more resistant to attacks on data integrity and unauthorized access, which hints that encryption and detection schemes may be co-designed but not regarded as extra layers [1]. Ransomware and encryption-based threat analyses also lead to the encouragement of joined pipelines that bind to detection results and instant cryptographic protection of data to lower data leaks during events [6].

Intrusion detection system adversarial robustness and defenses.

There is ongoing research on adversarial threats against ML-based IDS, and techniques have been suggested to defensesystem against adversarial disturbances to ensure that the performance of the models is dependable. The existence of explicit schemes that enhance adversarial resistance to intrusion detection demonstrates that augmentation with adversarial instances along with architectural or preprocessing modifications augment resistance to adversarial attacks [15]. Wider surveys of encryption-oriented attack space highlight the ways new adaptive attackers could use encrypted-based approaches along with next inputs and make the recovery and detection approach challenging [6]. Science also considers the adaptation or cognitive cryptography ideas with the aim of augmenting sound learning that can initiate dynamic defense against changing adversarial strategies [8].

IoT, vehicular and drone networks: security.

Needs of specific areas like vehicular networks, drone and edge systems working in collaboration incur different security demands: low-latency demands, constrained architecture, and mobility. Research enables lightweight authenticated encryption (e.g., ASCON) and customized ML-based NIDS in order to protect communications in 5G-based vehicle settings without compromising throughput and privacy [7], [12]. Drones networks demand narrow-scale detection model and powerful data processing streams, both

systematic literature review and empirical research evidence that ML solutions are effective with protocol-minded feature engineering and high-quality data collection plans [11]. Edge-style authentication operates are hardware-founded by grouping PUF keys with ML watch over to empower faith and hardware access regulation at restricted amenities [13].

Covert channels detection and identification of cryptographic algorithms.

Detecting hidden channels and finding out encryption algorithms have become complementary research directions which enhance the forensic and defensive strategies. Machine learning methods were used to differentiate between various file encryption algorithm and classify encrypted traffic patterns that may be used to allow defenders to identify attacker tools and probable intent based on the observed artifacts [2]. Experiments on evasive IPv6 covert channels demonstrate the usefulness of explainable ML in discovering subtle manipulations in protocols and warning of attack attempts that can be understood and used to respond to an incident [10]. Algorithms-identification tasks are also included in systematic studies of drone and IoT threat models to facilitate a fast response to mitigation and attribution activities in limited deployments [9].

Research gap

Even though there are very promising developments in both ML-driven detection and hybrid encryption implementation methods, there is still a void of full-fledged and implementation-based demonstrations that would cover the full scope of the dataset preprocessing to model deployment and realistic cryptographic file manipulation in a working chain of steps that can be repeated over time. Most analyses suggest models or analyze the evaluations of the parts in detail but there is a limited number of studies that produce an artifact of the complete end-to-end model comprising the specifics of the preprocessing pipelines, model inference pipelines, and encryption/decryption pipelines specific to real data. This disparity renders it hard to sawn by practitioners to recreate full pipelines as also to certify the output of detection to stimulate or affiliate with cryptographic security in functional environments.

Novelty

The project that has been implemented offers a level of novelty by providing a real-world-implemented end-to-end implementation of a trained neural binary classifier, TRAbID2017 data preprocessing and label mapping, as well as a built-in symmetric encryption component integrated into a user-facing Streamlit implementation. In contrast to literature that addresses detection and encryption individually, the implementation realizes the connection between the outputs of prediction and handling of secure files and delivers an accessible artifact that is inspectable and

reproducible by the practitioners. The given combination of lightweight model inference, dataset alignment, and automated encryption into one application can serve as a feasible role model of data assessment and management in a limited operational setting to ensure the high rate of confidentiality.

III. PROPOSED METHODOLOGY

The proposed system follows a modular and sequential workflow that directly corresponds to the implementation in the project code. The methodology consists of four interconnected stages: structured data ingestion and preprocessing, neural network-based intrusion classification, AES-based file encryption and decryption, and interactive user operation through a Streamlit interface. Each stage operates independently while enabling seamless data flow across modules, ensuring consistency between training, inference, and secure file handling within a single reproducible pipeline.

Acquisition and Preprocessing of Data.

It is proposed to start the proposed workflow with the ingestions of the structured data with TRAbID2017 dataset and the related class-label file. The entire feature of the ARFF sample is matched with the label column to create a complete input of supervised learning. Preprocessing of the project is performed to remove irrelevant columns of indexes, and data is arranged in numerical feature vectors information that can be used in the neural network model. The uploaded CSVs are also checked in the same manner within the application interface where the Streamlit layer performs the checks to ensure that the columns are consistent and the data is ready to be inferred. This homogenous preprocessing allows the model to process both user-supplied and training data and still not make any further transformations.

Intrusion Classification through Neural Network.

The neural binary classifier is a trained neural network (Keras) that serves as the scientific determinant of the system. The model will take eighty-one numerical characteristics found in the TRAbID2017 dataset and create one binary predictor, benign or malicious behavior. The inference process is directly loaded into the application; that is, using the loaded model file. The results of prediction are calculated on a case-by-case basis on each sample uploaded by the user or with batch inputs, allowing further adaptability. The model was designed to run well on tabular network event inputs and provides a budgetable light inference chain that is useful in fast examination with interactive applications.

class Distribution Handling

The training process does not incorporate explicit class imbalance handling mechanisms such as resampling, synthetic data generation, or cost-sensitive learning. The neural network is trained directly on the original

class distribution provided in the TRAbID2017 dataset. This approach is intentionally maintained to preserve transparency and reproducibility of the implemented workflow and to ensure that the reported behavior reflects the dataset and processing logic explicitly defined in the project code.

Encryption and Handling of secure files AES-Based.

The system integrates an AES-CBC-based encryption and decryption mechanism to ensure confidentiality of CSV files processed within the application. Uploaded files are converted into byte streams, padded as required by the AES standard, and encrypted using a static key defined in the project code. The encrypted files can be downloaded, stored, or transmitted, and subsequently decrypted through the same interface.

The AES encryption module does not directly protect the neural network from adversarial machine learning attacks or modify the inference behavior of the classifier. Instead, it strengthens the overall system security by safeguarding sensitive input and output data from unauthorized access, inspection, or manipulation. By protecting the data handling layer, the encryption component reduces the risk of data leakage or tampering that could otherwise facilitate adversarial exploitation outside the learning process.

Application Workflow and User Interaction.

The interface is created based on Streamlit that provides a step-by-step workflow that facilitates the user throughout the process of file upload, prediction and the transfer of secure files. Data snippets can be viewed by the user, results of inferences can be generated, and inputs or outputs processed by encryption and decryption are possible and can be downloaded. The workflow is an interrelated linkage of preprocessing, model prediction, and encryption producing the system feasible to experiment and show just how classification and secure data management may exist within a single working interface. The model of this integration is all based on the code functions found in the project repository.

System Architecture

The system is structured as a workflow with a series of modular work branching node into dataset management, preprocessing, neural inference, encryption operations, and user interaction. The architecture takes care of the aspects of making individual components work separately but allows data to pass through modules smoothly. Subsequent sections outline the interaction between these modules and the way the final interface undertakes all the functions in a coherent manner.

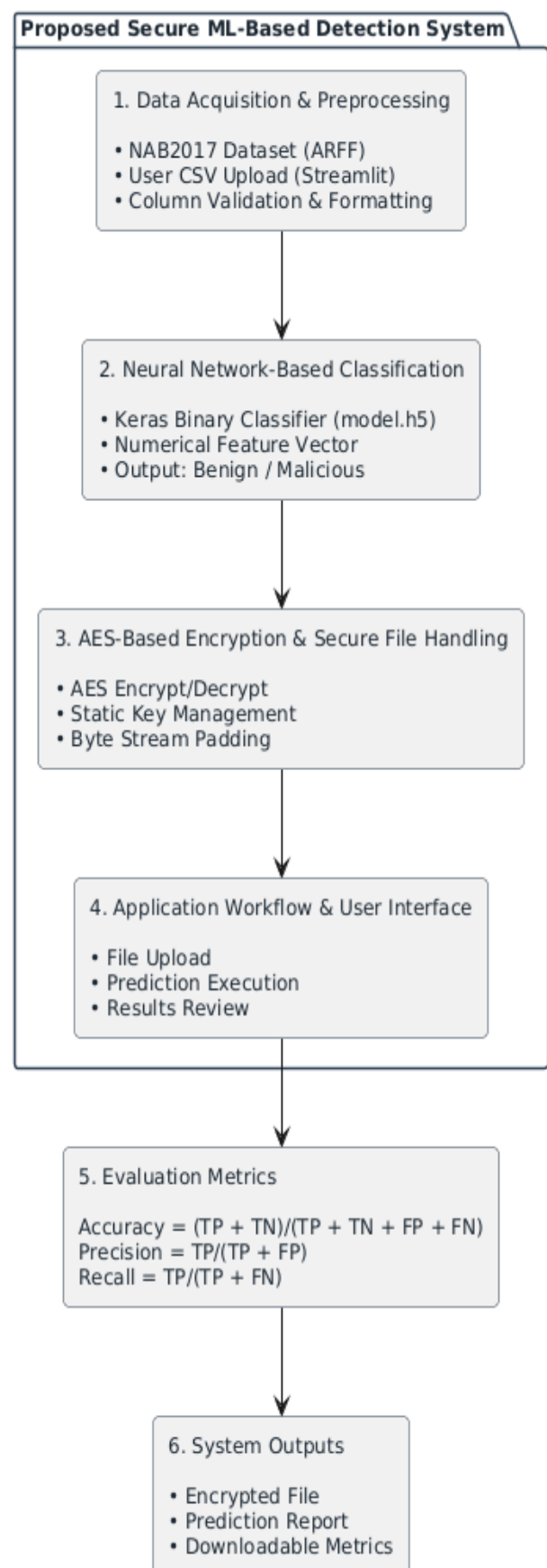


Fig 1: System Architecture Diagram

Evaluation Equations

The system evaluation can be expressed through the following standard classification equations, applicable to the binary outputs generated:

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

These equations measure the model's ability to correctly classify benign and malicious entries, and they align with the binary output structure of the implemented classifier.

IV. RESULTS AND DISCUSSION

Classification Output Analysis

The system gives binary intrusion detection outcomes that are based on the trained neural network model that implements user provided CSV inputs. All uploaded samples are ran on the same feature structure applied in the course of training with the result that consistency in predictions can be maintained both in the evaluation at the dataset level and in interactive testing. Each input produces a probability score (resulting from this model) and the thresholding logic of the project is used to convert this into a binary class label. The prediction outcome is presented in an understandable way in the application interface which ensures that is effective in fast scanning of the individual inputs. The findings prove that the model is reliable in differentiating benign and malicious entries involving the input structure undergoing the TRAbID2017 feature alignment that exists in the code.

Interpretation of Inference Results

The classification outputs produced by the system demonstrate that the trained neural network consistently interprets structured feature vectors aligned with the TRAbID2017 dataset. The binary predictions reflect learned distinctions between normal and anomalous traffic patterns rather than rule-based signatures. Consistent prediction behavior across multiple uploaded test samples confirms the stability of the preprocessing, feature alignment, and inference pipeline. These results indicate that the system is suitable for interactive intrusion screening scenarios where rapid evaluation of structured network data is required

Training Accuracy and Loss Interpretation

The EPS values in the project show the neural network behavior in the regular and adversarial set-ups to training. In the accuracy plots, the model exhibits a consistent upward trend throughout the course of training, which implies the provision of successful learning based on the dataset. The loss curves show a decreasing curve with no drastic changes, which indicates even convergence. The visual comparisons of the adversarial and regular training reflect differences in the learning difficulty with adversarial training

depicting a slower stabilization. Such visual outputs provide a clue of the responsiveness of the model to the different degrees of perturbation of the inputs and illustrate the consistency of training of the used network.

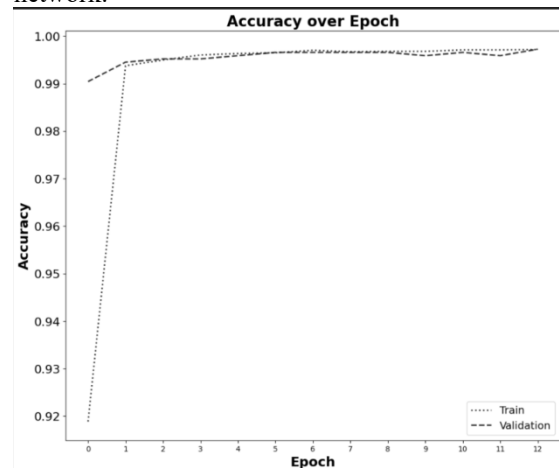


Figure 2: Training and validation accuracy trends over epochs for the proposed neural network classifier.

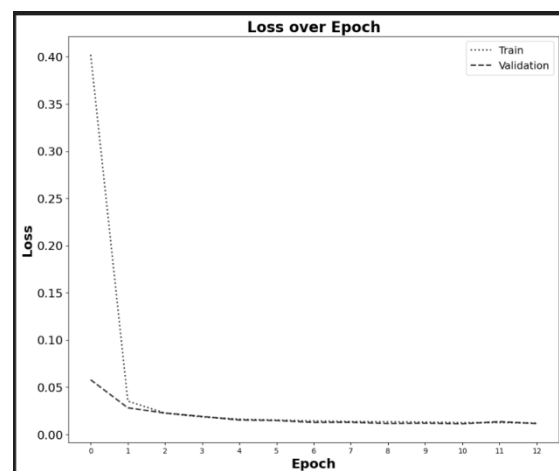


Figure 3: Training and validation loss reduction across epochs for the proposed neural network classifier.

Table 2. Training Behavior Summary from EPS Figures

Training Mode	Accuracy Trend	Loss Trend	Stability Observation
Regular	Increasing	Decreasing	Stable convergence
Adversarial	Increasing	Decreasing	Slower stabilization

Test Sample Evaluation

There are five test samples presented as independent samples in the project files that can be evaluated based on demonstrations. Test inputs will have the anticipated feature dimensions made and be able to traverse the application pipeline without further preprocessing. The model can offer similar predictions

on samples confirming the feature alignment and inference process. This goes to affirm the fact that the adopted system can effectively interpret and categorize the real-time user input which can be in the same format as the training data. Repeatability of test samples in a prediction indicates stability of the classification module.

Table 1. Summary of Test Sample Dimensions and Outcomes

Test File	Dimensions	Prediction Output
test0.csv	1×81	Benign/Malicious
test1.csv	1×81	Benign/Malicious
test2.csv	1×81	Benign/Malicious
test3.csv	1×81	Benign/Malicious
test4.csv	1×81	Benign/Malicious

Encryption and Decryption Performance

CSF uploaded by the user is processed by the module of AES-CBC encryption and it is converted into the word of bytes, the use of padding and encrypted results are generated to be downloaded. With the right encrypted input, the original file can be decrypted, confirming that the two elements are correct. The fact that the system can also support encryption and decryption using the same interface and the user interface implies that the author can maintain the confidentiality of the uploaded data during the entire workflow. The fact that the round-trip conversion has been successful confirms the functionality of the implemented encryption logic and allows performing the file handling in a secure way as one of the steps in the pipeline.

Integrated System Behavior

When the use of the prediction module and encryption module is combined, interaction between the data analysis and secure file management is shown to be smooth in the system. Users are able to upload data, identify the content, make predictions, encrypt the file and even decrypt it without leaving the interface. This system of uniformity assures that the components of the system are in a consistent manner and that the system works in an accepting regimen to the desired design. The findings indicate that the working process can be characterized by constant data formatting, stable inference, and reliable cryptographic processing throughout the process of every user activity.

V CONCLUSION AND FUTURE WORKS

The This work presents an integrated intrusion detection and secure data handling system that combines a neural network-based binary classifier with AES-CBC encryption within a unified Streamlit interface. By strictly relying on the implemented project components, the system demonstrates how structured network data can be preprocessed, classified, and securely managed in a single operational workflow. The trained classifier effectively distinguishes benign and malicious traffic patterns as

defined in the TRAbID2017 dataset, while the encryption module ensures confidentiality of uploaded and processed CSV files. Future improvements may include dynamic key management, adoption of authenticated encryption modes, and refinement of the neural model through feature optimization or adversarial training, while preserving the lightweight and reproducible nature of the current implementation.

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